



PM Plan: Barangay Disaster Preparedness and Response System

Prepared for

IT317 – PROJECT MANAGEMENT

Submitted to

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BantayKalikasan: Barangay Disaster Preparedness and Response System

Project Objective

Project Name:

BantayKalikasan: Barangay Disaster Preparedness and Response System

To develop a centralized web-based platform that enables barangay officials and residents to prepare for, respond to, and recover from natural disasters more effectively through alerts, evacuation planning, resource tracking, and community engagement tools.

Vision

Create resilient barangays equipped with digital tools to reduce disaster risk and improve emergency response.

Mission

Build an accessible, real-time system to coordinate disaster information, manage response logistics, and empower residents with lifesaving knowledge.

Core Deliverables

- **Disaster Alert System**
Push real-time weather or disaster alerts from DOST-PAGASA or PHIVOLCS to residents.
- **Evacuation Plan Map**
Show evacuation centers, safe zones, and routes.
- **Barangay Resource Tracker**
Inventory of supplies (e.g., food, meds, rescue gear) and personnel.
- **Resident Check-in Portal**
Residents can mark themselves safe or request help.
- **Admin Dashboard**
Update maps, add resources, send alerts, generate reports.

Problem Statement

Core Issue:

Many barangays in the Philippines lack a digital system to manage disasters, resulting in delayed responses, miscommunication, and inefficient relief distribution.

Proposed Solution:

A centralized platform to disseminate disaster alerts, monitor evacuation progress, track supplies, and enable citizen check-ins to streamline emergency response and save lives.

Supporting Data:

- According to NDRRMC, over 60% of barangays lack structured disaster response systems.
- PAGASA reports an increase in the frequency of typhoons due to climate change.
- Digital early warning systems can reduce disaster casualties by up to 40% (UNDRR, 2023).

Scope Statement

Scope (Inclusions):

- Responsive web application for residents and barangay staff
- Real-time notifications and alerts
- Map integration for evacuation planning
- Supply inventory and personnel tracking
- Admin tools for monitoring and reporting

Exclusions:

- Mobile offline access
- National-level system integration
- SMS alert system

Constraints:

- 4-month timeline
- PHP 950,000 budget
- Dependent on internet connectivity

Assumptions:

- Residents have basic internet access
- Barangay officials will participate in training
- LGU will support implementation

Stakeholder Analysis

Stakeholder	Role	Interest	Influence
Project Team	Developers/P M	Successful delivery	High
Barangay Officials	Admin Users	Improve disaster management	High
Residents	End Users	Receive alerts and guidance	High
LGU Partners	Support	Funding and oversight	Medium
DRRM Volunteers	Field Responders	Coordinate evacuation and relief efforts	Medi

Work Breakdown Structure (WBS)

1. Initiation
 - a. Identify stakeholders
 - b. Define project charter
2. Planning
 - a. Define requirements and risk assessment
 - b. Design mockups and system architecture
3. Execution
 - a. Develop modules: alerts, maps, inventory, check-in
 - b. Integrate mapping and database systems
4. Monitoring & Control
 - a. Sprint reviews
 - b. Bug tracking and internal testing
5. Closure
 - a. Final deployment and training
 - b. Post-deployment support plan

Schedule and Key Milestones

Milestone	Timeline	Description
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Project Kickoff	Week 1	Finalize scope and responsibilities
Mockups Completed	Week 3	Approve system design and UI
Core Development	Weeks 4–10	Build and integrate main modules
Testing & Adjustments	Weeks 11–13	UAT and security testing
System Launch	Week 14	Go live and conduct community onboarding

Cost Estimate

Category	Estimated Cost (PHP)	Description
Development	600,000	Full-stack programming & integrations
Hosting & Tools	120,000	Servers, APIs, map services
Training & Materials	80,000	Barangay seminars and guides
Contingency & Legal	150,000	Backups, documents, legal support
Total	950,000	

Key Adjustments and Budget Optimization

- Use open-source map tools like Leaflet.js
- Host prototype on Netlify or Firebase
- Conduct digital onboarding to save on materials

Risk Analysis

Risk	Likelihood	Impact	Mitigation
Internet Downtime	Medium	High	Design fallback cache system
Low User Adoption	Medium	Medium	Conduct awareness seminars
Data Breach	Low	High	SSL, hashed login, routine audits

Communication Plan

Weekly & Monthly Communications:

- Slack for daily updates
- Google Meet for weekly stand-ups
- Monthly reports shared via Google Docs

Expanded Communication Plan:

Communication Type	Audience	Frequency	Mode	Owner
Dev Progress	Team Members	Weekly	Slack, Meet	Project Lead
Stakeholder Rdeports	LGU/Officials	Monthly	Docs, Email	PM
Awareness Updates	Residents	As Needed	Facebook, SMS	Outreach Lea

Team Roles and Responsibilities

- **Project Manager:** Oversees timeline and quality
- **Frontend Developer:** UI/UX development
- **Backend Developer:** API, database, and logic
- **QA Tester:** Test automation and manual validation
- **Trainer/Support:** Conducts seminars, assists users

Testing and Validation

Testing Types:

- Unit Testing: Validate modules like alerts, login
- Integration Testing: Ensure data sync between UI and DB
- UAT: Trial with barangay officials and residents

Tools and Methods:

- Jest, Cypress for frontend
- Postman for API testing
- Manual testing with sample data

Project Tools and Resources

- **Project Management:** Trello, GanttProject
- **Development Stack:** React.js, Firebase, Leaflet.js
- **Version Control:** GitHub
- **Comms & Docs:** Slack, Google Drive, Canva

Mockups and System Wireframes

- **Alert Dashboard:** Admin can publish alerts, view history
- **Evacuation Map Interface:** Highlight routes and centers
- **Resource Inventory Tracker:** Live tracking of supplies and responders

Success Metrics

Metric	Target Value
Resident Registrations	500 in 3 months
Alert Reach Rate	90% within 10 minutes
System Uptime	99%
User Satisfaction	4.5/5 average rating

Change Management Plan

- Log all change requests
- Review for impact and feasibility
- Discuss with stakeholders
- Update documentation and scope accordingly

Security and Data Protection Measures

- Enforce HTTPS and TLS encryption
- Two-factor authentication for admins
- Weekly encrypted data backups
- Conduct periodic vulnerability scans

Sustainability and Future Expansion

- Build mobile-friendly version
- Add chatbot for emergency FAQs
- API integration with LGU and NDRRMC data feeds
- Enable volunteer coordination system

Contingency Plan

Scenario	Contingency Action
System Downtime	Shift to backup host (e.g., Firebase)
Low Engagement	Roll out barangay-level training
Data Loss	Restore from encrypted weekly backup

Conclusion

BantayKalikasan aims to create a digital-first approach to disaster risk management at the barangay level. By combining alert systems, evacuation planning, and real-time communication in a unified platform, this project empowers communities to be more resilient, responsive, and prepared.

21. References

- <https://ndrrmc.gov.ph/>
- <https://www.pagasa.dost.gov.ph/>
- <https://www.unisdr.org/>
- https://en.wikipedia.org/wiki/Project_Management_Body_of_Knowledge
- <https://www.dilg.gov.ph/>